

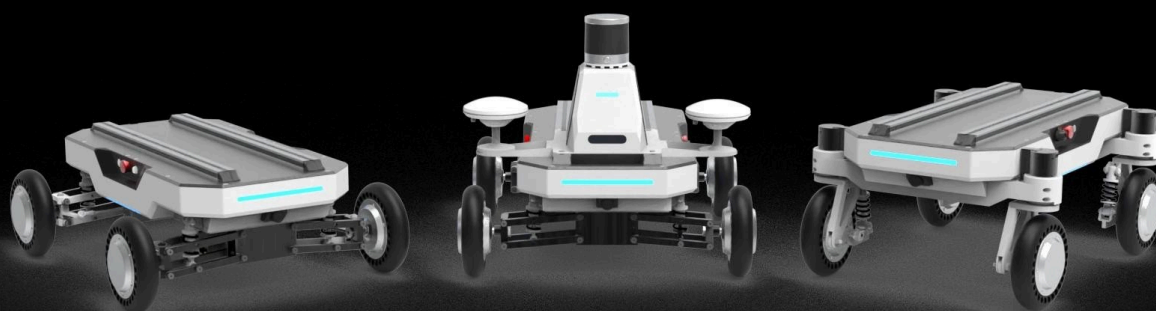
- Four-wheel-steer robot chassis

LUCHS A

**BUILT FOR ROBOTS.
STEERS EVERY WHEEL.**

A compact, open mobile base: up to 80 kg on board, four drive configurations from skid-steer to full four-wheel steering, and the same CAN interface as the rest of the Luchs family.

CHASSIS BY RJ TECH LIMITED / AUTONOMY BY TERRAAGENT ROBOTICS (TAR)



Shown: dual Ackermann · with optional sensor mast · four-wheel steering

80 kg

Maximum payload

11.6 km/h

Top speed

0 m

Turning radius: pivots on the spot

6 h

Runtime unloaded; 3–5 h loaded

48 V

LiFePO₄ battery, 2 h full charge

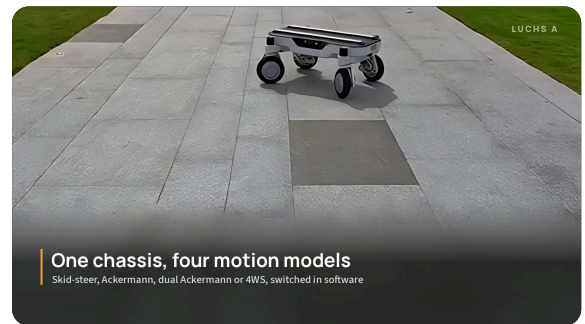
15 °

Maximum gradient

A COMPACT, OPEN MOBILE BASE FOR ROBOTS THAT NEED TO GET AROUND.

Luchs A gives your robot a reliable pair of legs. It carries up to 80 kg, steers all four wheels, and talks to your computer over standard industrial interfaces. Mount your sensors and payload on the top rails, plug into the expansion bay, and start building the application instead of the drivetrain.

- **Four drive configurations, one platform.** Skid-steer, front Ackermann, dual Ackermann or full four-wheel steering. The main body is shared; only the drive module changes.
- **Open by design.** CAN, RS-485 and RS-232 control, a published protocol, and ROS 2 software with an SDK, robot model and simulator in the box.
- **Power for your payload.** 48 V / 10 A and 12 V / 8 A outputs, each with over-current, short-circuit and over-temperature protection.
- **Safety you don't have to add later.** Dual emergency stops, ultrasonic collision stop, power-on self-test and a monitored battery.



Luchs A-4WS on the move: crab walk, slalom, pivot turns and an 80 kg ride. Watch the film: [Luchs_A_Promo_EN_1080p.mp4](#).



PICK THE DRIVETRAIN THAT FITS THE JOB.

Every Luchs A shares the same main body, electronics, interfaces and safety system. The drive module underneath decides how it moves.

SIMPLEST

Luchs A-DF

Differential 4WD (skid-steer)

Size (LxWxH)

800x665x292

mm

Weight

40 kg

Steering motors

None

Turning radius

0 m (pivot)

CAR-LIKE

Luchs A-FA

Front-Ackermann 4WD

Size (LxWxH)

800x665x292

mm

Weight

45 kg

Steering motors

1 x 140 W

Turning radius

0.89 m · 0 m

pivot

TIGHT CAR-LIKE

Luchs A-DA

Dual-Ackermann 4WD

Size (LxWxH)

800x665x292

mm

Weight

50 kg

Steering motors

1 x 140 W

Turning radius

0.445 m · 0 m

pivot

FLAGSHIP

Luchs A-4WS

Four-wheel independent steering, 4WD

Size (LxWxH)

800x580x372

mm

Weight

55 kg

Steering motors

4 x 140 W

Turning radius

0 m · moves

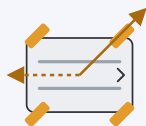
sideways

All four configurations: 4 x 250 W hub drive motors, 206 mm wheels, 594 mm wheelbase, 80 kg payload, 0–11.6 km/h.

MOVES THE WAY YOUR TASK NEEDS.



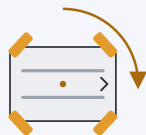
The A-4WS supports all four motion models below and switches between them on a single command. Your planner can use whichever model suits it, and the reported odometry follows the model you selected.



Crab walk

All four wheels point the same way, so the chassis slides sideways or diagonally without turning its nose.

A-4WS



Pivot turn

The chassis spins about its own centre. Ideal for narrow aisles and docking.

ALL CONFIGURATIONS



Skid-steer

Wheels stay straight; the chassis turns by driving the left and right sides at different speeds.

ALL CONFIGURATIONS



Ackermann

Front wheels steer like a car. Smooth, predictable arcs that are gentle on tyres and floors.

A-FA · A-DA · A-4WS



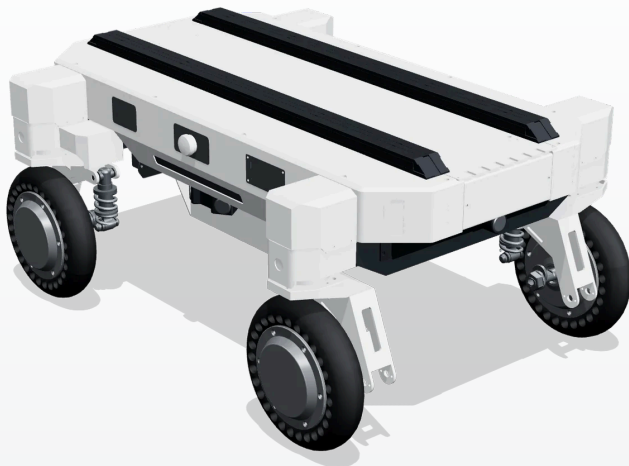
Dual Ackermann

Front and rear axles steer in opposite directions, halving the turning radius.

A-DA · A-4WS

SIMPLE OUTSIDE. SERVICEABLE INSIDE.

A carbon-steel and aluminium frame with independent suspension at every corner. The A-4WS sits on swing-arm suspension with 220 mm of ground clearance; the other configurations use double wishbones front and rear.



Luchs A-4WS, IP54-sealed build. CAD render.



Side view: power button, mode selector and charging port are all within reach.



Rear three-quarter view. Two 2040 aluminium rails run the full length of the deck.

THE AUTONOMY IS OURS. BUILT BY TAR, FOR THE LUCHS FAMILY.

RJ Tech builds the chassis. TerraAgent Robotics (TAR) turns it into a robot that does your job: we integrate the sensors and the computer, set up positioning and navigation, and adapt the software to your site. The building blocks are shared across the Luchs family and already run on real vehicles, on real data.

01 WHAT WE DELIVER

Navigation kit

Built-in computer with ROS pre-installed, LiDAR-based SLAM (GMapping and Cartographer) and the ROS navigation stack.

- Live mapping, online and offline
- Global and local path planning
- Dynamic obstacle avoidance and trajectory tracking

Sensor integration

We mount, wire and calibrate the sensors your application needs, using the deck rails and the protected power outputs.

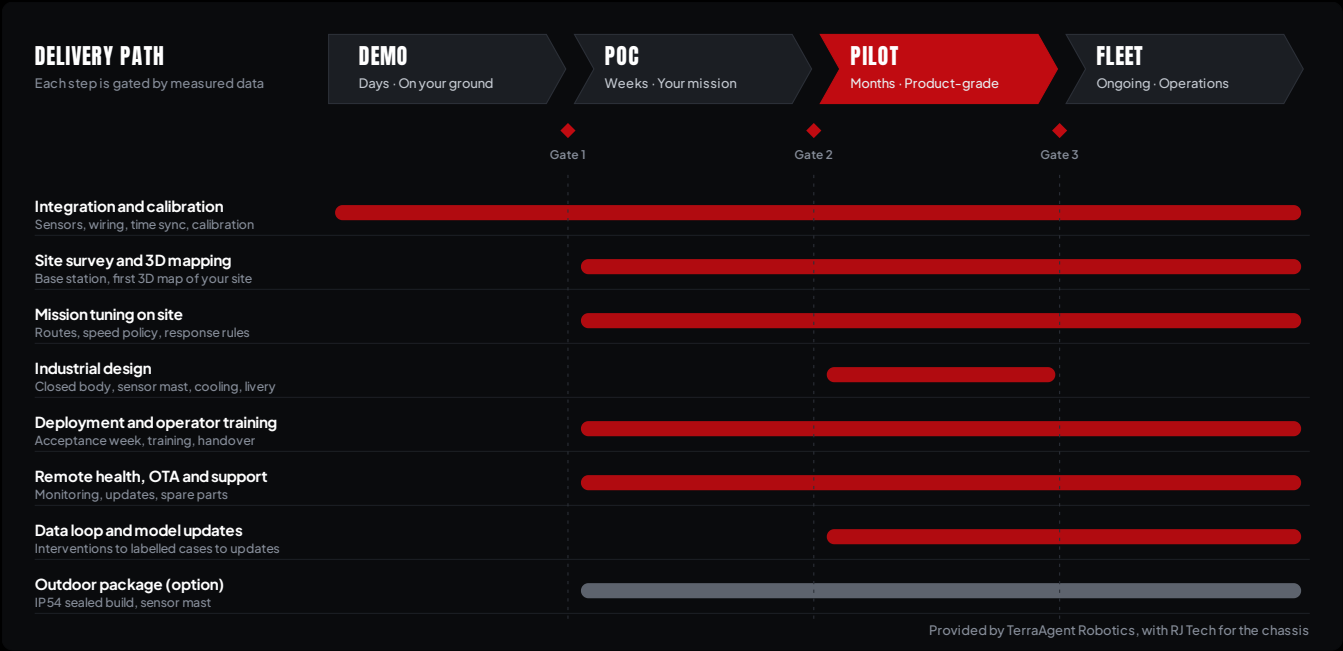
- Single-line LiDAR and depth cameras
- IMU and wheel-encoder fusion
- Custom masts and payload brackets

Application engineering

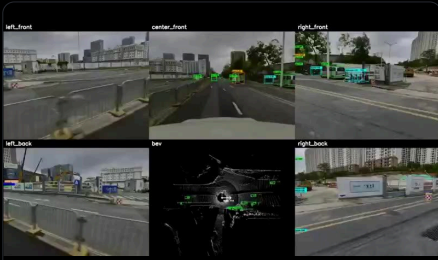
From a first demo to a pilot fleet: we help you define the task, adapt the software, and test it where the robot will actually work.

- Mission setup and tuning
- Data logging and visualisation
- Training and handover

02 BEYOND THE SOFTWARE



03 ALREADY RUNNING ON REAL VEHICLES AND REAL DATA



Camera-first perception

Five cameras and LiDAR fused into one bird's-eye view with 3D detection, projected back into every camera.

REAL VEHICLE



Dense 3D reconstruction

Three front cameras build a coloured 3D model of the road as the vehicle drives. No LiDAR.

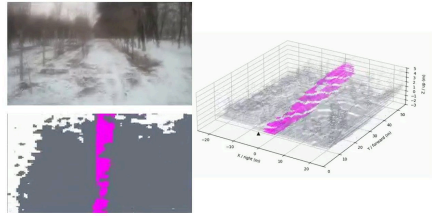
REAL VEHICLE



Reconstruction against low sun

The same reconstruction on a suburban road against the sun; the top view keeps the street shape through the turn.

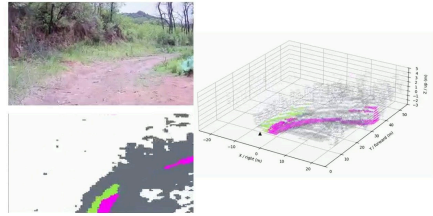
REAL VEHICLE



Drivable ground in snow

One camera, a 3D occupancy grid and a drivable-ground map (magenta) on a snow-covered forest track.

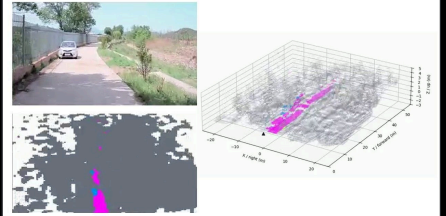
RECORDED OFF-ROAD DATA, PROCESSED OFFLINE



Dirt trail and verge

Follows a curving dirt track through the bends and tells the track from the verge (green).

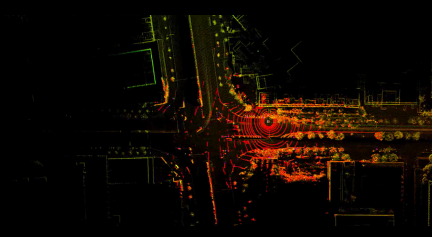
RECORDED OFF-ROAD DATA, PROCESSED OFFLINE



Oncoming vehicle

A narrow lane along a fence: the space taken by the oncoming car is cut out of the drivable ground.

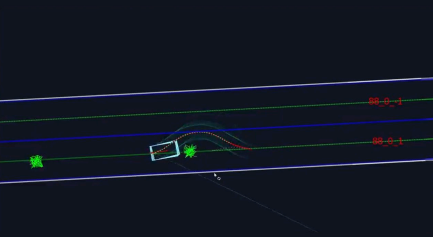
RECORDED OFF-ROAD DATA, PROCESSED OFFLINE



LiDAR map localisation

Each LiDAR scan registered to a prebuilt point-cloud map at 20 Hz: the second source of position when satellites drop out.

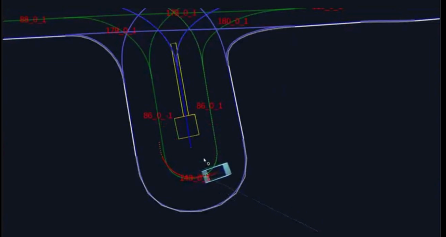
REAL VEHICLE, REAL TIME



Planning around an obstacle

The planner leaves a blocked lane, passes the obstacle and returns to the lane on a smooth path.

PLANNING STACK, REPLAY



Junctions, U-turns and parking

Route, behaviour and trajectory in three layers: turns at junctions, a U-turn and reversing into a bay.

PLANNING STACK, REPLAY

Real-vehicle tiles show our own software running on real vehicles. Off-road tiles are our perception model run offline on recorded off-road driving data. Planning tiles are replays of our planning stack. All are building blocks we integrate on the Luchs family.

BOLT IT ON. PLUG IT IN. DRIVE IT FROM CODE.



Mounting rails

Two 2040 aluminium extrusions, 700 mm long at 200 mm centres, take standard 40-series slot nuts and brackets. Mount sensor masts, robot arms, lifting modules or cargo boxes without drilling the chassis.



Expansion bay

Under a magnetic cover at the rear of the deck: one DB9 signal connector (CAN, RS-485, RS-232) and one power connector with 48 V / 10 A and 12 V / 8 A outputs.



External or built-in computer

Connect your own industrial PC or edge computer to the expansion bay, or order the built-in computer with HDMI, USB 2.0 and Ethernet ports on the chassis.



Software that's ready on day one

The ROS 2 driver takes `cmd_vel` and publishes odometry, joint states, chassis and safety state, battery data and diagnostics, with watchdogs and limits built in. A C++ SDK, a command-line tool and a built-in simulator let you start before the chassis arrives. The full CAN and serial protocols are documented if you prefer to write your own.

Interfaces at a glance

INTERFACE	DETAILS
CAN	CAN 2.0B, 1 Mbit/s, standard 11-bit frames, 8-byte payload
RS-485	Fixed-length frames with header and tail bytes
RS-232	Same frame format as RS-485
Power out	48 V / 10 A and 12 V / 8 A, individually protected
Built-in computer (option)	HDMI, USB 2.0 Type-A, RJ45 Ethernet
Remote control	10-channel 2.4 GHz transmitter and receiver included

```
# Command 0.5 m/s forward on the A-4WS (CAN ID 0x5A0)
vx = 500 # 0.001 m/s per bit → bytes 0-1
vy = 0   # lateral, A-4WS only → bytes 2-3
wz = 0   # 0.001 rad/s per bit → bytes 4-5
model = 0x03 # four-wheel steering → byte 6
# → 01 F4 00 00 00 00 03 00
```

LAYERED PROTECTION, BUILT IN.



Safety on Luchs A doesn't depend on your software being right. The chassis watches itself, and the most important protections work at the hardware level.

Two emergency stops

One on each side. A press cuts motion commands, brakes the motors and locks the chassis until it is reset.

Ultrasonic collision stop

In remote-control mode the chassis brakes if an obstacle comes within 20 cm. In automatic mode the ranges are reported to your software.

Power-on self-test

Checks the battery, power rails, communication links, drive motors and ultrasonic sensors before the chassis will move.

Battery management

The BMS protects against over-charge, over-discharge, over-current and abnormal temperature.

Charging interlock

With the charger plugged in, the drive is disabled and the chassis ignores all motion commands.

Protected outputs

Every power channel is current-monitored with software cut-off; critical outputs add hardware protection against overload, short circuit and over-temperature.

● Remote control

The handheld transmitter drives the chassis. It can hand control to your computer at the flick of a switch, and take it back just as quickly.

● Maintenance

Motion is locked out. Used to clear faults, reset after an emergency stop, or release the brakes to push the chassis by hand.

● Automatic

Your computer drives over CAN, RS-485 or RS-232. The light bar turns blue so everyone nearby knows who's in charge.

TECHNICAL DATA

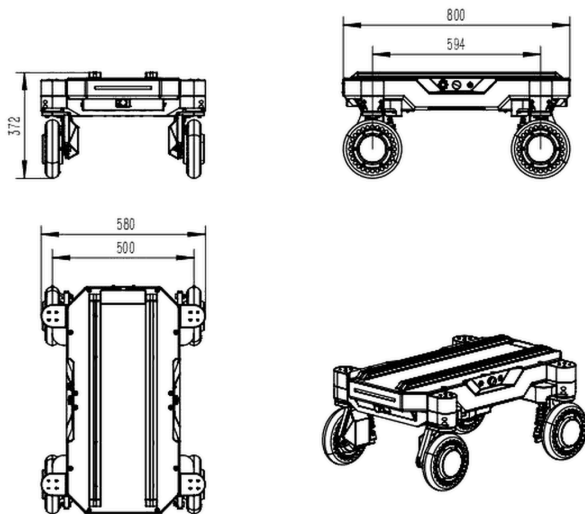


	A-DF Differential 4WD	A-FA Front-Ackermann 4WD	A-DA Dual-Ackermann 4WD	A-4WS Four-wheel steering
MOTION				
Special manoeuvres	Pivot turn, tight U-turn	Pivot turn, tight U-turn	Pivot turn, tight U-turn	Pivot turn, omnidirectional travel
Motion models	Skid-steer	Skid-steer, Ackermann	Skid-steer, Ackermann, dual Ackermann	Skid-steer, Ackermann, dual Ackermann, 4WS
Speed	0–11.6 km/h			
Minimum turning radius	0 m (pivot)	0 m (pivot) · 0.89 m (Ackermann)	0 m (pivot) · 0.445 m (dual Ackermann)	0 m
Pivot-turn clearance circle	1.01 m	1.01 m	1.01 m	0.857 m
MECHANICAL				
Dimensions (L×W×H)	800×665×292 mm	800×665×292 mm	800×665×292 mm	800×580×372 mm
Wheelbase	594 mm			
Track width	596 mm	596 mm	596 mm	500 mm
Wheel diameter	206 mm			
Ground clearance	77 mm	77 mm	77 mm	220 mm
Suspension	Double wishbone, front and rear	Double wishbone, front and rear	Double wishbone, front and rear	Independent swing arm
Chassis weight	40 kg	45 kg	50 kg	55 kg
Frame material	Carbon steel and aluminium alloy			
Mounting rails	2 × 2040 aluminium extrusion, 700 mm long, 200 mm centres			
PERFORMANCE				
Maximum payload	80 kg			
Maximum gradient	15°			
Parking brake	Electric motor hold			
Runtime	Up to 6 h unloaded · 3–5 h loaded			
Operating temperature	–10 °C to +45 °C			
Ingress protection	IP33 (standard build) · IP54–sealed A-4WS build available			
DRIVE AND POWER				
Drive motors	4 × 250 W servo hub motors			
Steering motors	None	1 × 140 W servo	1 × 140 W servo	4 × 140 W servo
Battery	LiFePO ₄ , 48 V, 15 Ah (720 Wh)			
Charging	48 V / 10 A charger included · 0–100 % in about 2 h			
Power outputs	48 V / 10 A × 1 · 12 V / 8 A × 1			
CONTROL AND COMMUNICATION				
Operating modes	Remote control · Automatic · Maintenance			
Remote control	10-channel 2.4 GHz transmitter with receiver			
Communication	CAN × 1 · RS-485 × 1 · RS-232 × 1			
Software	ROS 2 Humble / Jazzy and ROS 1 Noetic (luchs_ros 1.1: SDK, driver, robot model, simulator; one code base with Luchs B) · CAN and serial protocol documentation · DBC files			
Lighting	Status lights, breathing light, warning lights, charge-progress bar			
SAFETY				
Emergency stop	Two buttons, left and right			
Ultrasonic sensors	2 (front, rear)	2 (front, rear)	2 (front, rear)	6 (1 front, 1 rear, 2 per side)
Protection	Power-on self-test and fault warning · BMS · charging over-current · battery surge · actuator over-current			

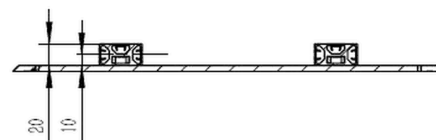
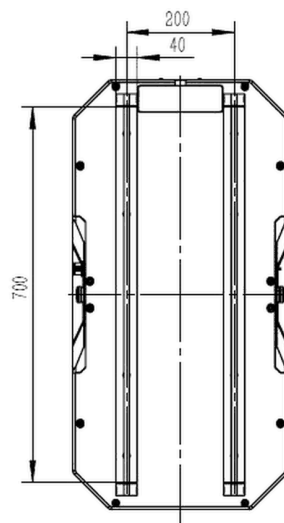
Typical values. Speed, runtime and gradeability depend on payload, surface and driving style. The pivot-turn clearance circle is the diameter the chassis sweeps when it spins on the spot. Specifications may change without notice. Other wheel, battery and enclosure options are available on request.

DRAWINGS

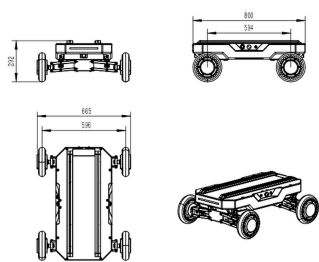
All dimensions in millimetres.



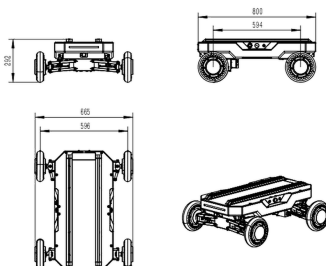
Luchs A-4WS, standard build: 800 × 580 × 372 mm, wheelbase 594 mm, track 500 mm.



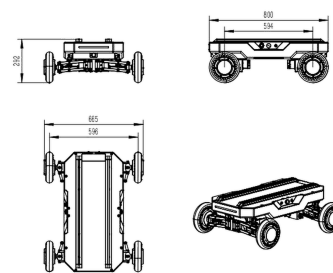
Top deck: two 2040 rails, 200 mm centres, 700 mm long.



A-DF: 800 × 665 × 292 mm



A-FA: 800 × 665 × 292 mm



A-DA: 800 × 665 × 292 mm

EVERYTHING YOU NEED TO START.



User manual

Safety, quick start, controls, safety functions, CAN and serial protocols, ROS software, TAR autonomy.

Luchs_A_User_Manual_EN_v1.4_20261005.pdf

Product brochure

This page as a printable PDF.

Luchs_A_Brochure_EN_v2.3_20261005.pdf

Data sheet

Three-page technical summary with the family line-up.

Luchs_A_DataSheet_EN_v1.0_20261005.pdf

CAN protocol

All messages, signals, scaling and error codes, with DBC files.

Luchs_A_CAN_Protocol_v1.1.xlsx · Luchs_A_CAN_v1.1.dbc

Serial protocol

RS-485 / RS-232 frames for each configuration.

Luchs_A_Serial_Protocol_v1.0.xlsx

ROS2 software

SDK, driver, robot model, launch files, tools and documentation. Humble and Jazzy. One code base for Luchs A and Luchs B.

luchs_ros2_v1.1.0.zip

ROS1 software

The same SDK and driver for ROS1 Noetic.

luchs_ros1_v1.1.0.zip

3D model

Luchs A-4WS, IP54 build: STEP for CAD, GLB for viewers.

Luchs_A-4WS_IP54_STEP.zip · Luchs_A-4WS_IP54.glb

Product film

About 40 seconds, 1080p and 720p.

Luchs_A_Promo_EN_1080p_v2.0_20261005.mp4

TERMS AND ABBREVIATIONS



4WD Four-wheel drive: every wheel has its own drive motor.

4WS Four-wheel steering: every wheel can be steered on its own.

Ackermann steering Car-style steering in which the inner wheel turns more sharply than the outer one, so both follow the same arc.

BMS Battery Management System: electronics that monitor and protect the battery.

CAN Controller Area Network: a robust two-wire bus widely used in vehicles and machines.

Crab walk Sideways or diagonal travel with all wheels pointing the same way.

DB9 A 9-pin D-subminiature connector.

IP33 / IP54 Ingress Protection rating to IEC 60529. IP33: protected against objects over 2.5 mm and spraying water. IP54: dust-protected and protected against splashing water.

LiFePO₄ Lithium iron phosphate: a lithium-ion chemistry known for long cycle life and thermal stability.

Odometry An estimate of the robot's motion computed from wheel speeds and steering angles.

Pivot turn Turning on the spot about the chassis centre; turning radius 0 m.

ROS Robot Operating System: the most widely used open-source robotics middleware.

RS-232 / RS-485 Serial communication standards. RS-485 is differential and suited to longer, noisier cable runs.

Skid-steer Steering by driving the left and right wheels at different speeds; also called differential drive.

SLAM Simultaneous Localisation and Mapping: building a map while working out where the robot is on it.

• Contact

ONE FAMILY. TALK TO US.

Luchs A is the most versatile member of the Luchs family; Luchs B is built for harder outdoor duty. One interface, one software stack. Tell us your application, and we will configure the chassis and the autonomy for it.

	Luchs A this product	Luchs B full-size · built for harder duty	Luchs C custom
Built for	Robots that need many motion modes and interfaces, indoors and out	Off-road and site work: speed, range, clearance, heat and dust	Your specification
External dimensions	800×580×372 mm (A-4WS) 800×665×292 mm (other builds)	1990×1150×800 mm	Customisable
Chassis weight	40–55 kg	230 kg	Customisable
Working / max. payload	— / 80 kg	80 kg / 200 kg	Customisable
Max. climb grade	15°	30°	30°
Ground clearance	77 mm / 220 mm (A-4WS)	300 mm	Customisable
Steering	Skid-steer, Ackermann, dual Ackermann or four-wheel steer	Single or dual Ackermann (dual EPS)	Customisable
Max. speed	11.6 km/h	40 km/h	55 km/h
Drive	4 × 250 W hub motors	4 × in-wheel motors, 5 kW peak each	260 N·m×4 / 330 N·m×4
Battery	LiFePO ₄ , 48 V 15 Ah	LiFePO ₄ , 72 V 60 Ah / 73 V 127.4 Ah	LiFePO ₄ , up to 96 V 130 Ah
Endurance	Up to 6 h (no load)	≥60 km (full load), up to 300 km in desert mode	≥60 km, up to 300 km with range extender
Ingress protection	IP33 (IP54 build available)	IP67 (≤440 mm) / IP65 (>440 mm)	IP67
Host interface	CAN 1 Mbit/s (Luchs A protocol v1.1) · RS-485 / RS-232	CAN 1 Mbit/s, LCI v1.0 with heartbeat and live limits	LCI over CAN / CAN FD
Software	ROS 2 · ROS 1 · C++ SDK	ROS 2 · ROS 1 · C++ SDK	ROS 2 · ROS 1 · C++ SDK

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RJTech

LUCHS A

Four-wheel-steer robot chassis

CHASSIS

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AUTONOMY

TerraAgent Robotics (TAR)
Contact: Brandon Zhe Feng

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